

MedGemma

Chest X-ray

Training Companion Proposal

An AI-assisted student radiology trainer using MedGemma and Retrieval-Augmented Generation

PRESENTED BY

Minura Wijewickrama

Introduction to Chest X-ray Training

- CHEST X-RAY INTERPRETATION IS A CORE RADIOLOGY SKILL
- STUDENTS NEED REPEATED PRACTICE WITH STRUCTURED FEEDBACK
- AI CAN SUPPORT LEARNING BY EXPLAINING FINDINGS AND EVIDENCE
- THE SYSTEM IS DESIGNED AS A TRAINING COMPANION, NOT A CLINICAL DECISION TOOL

Student-in-the-Loop Chest X-ray Training Companion

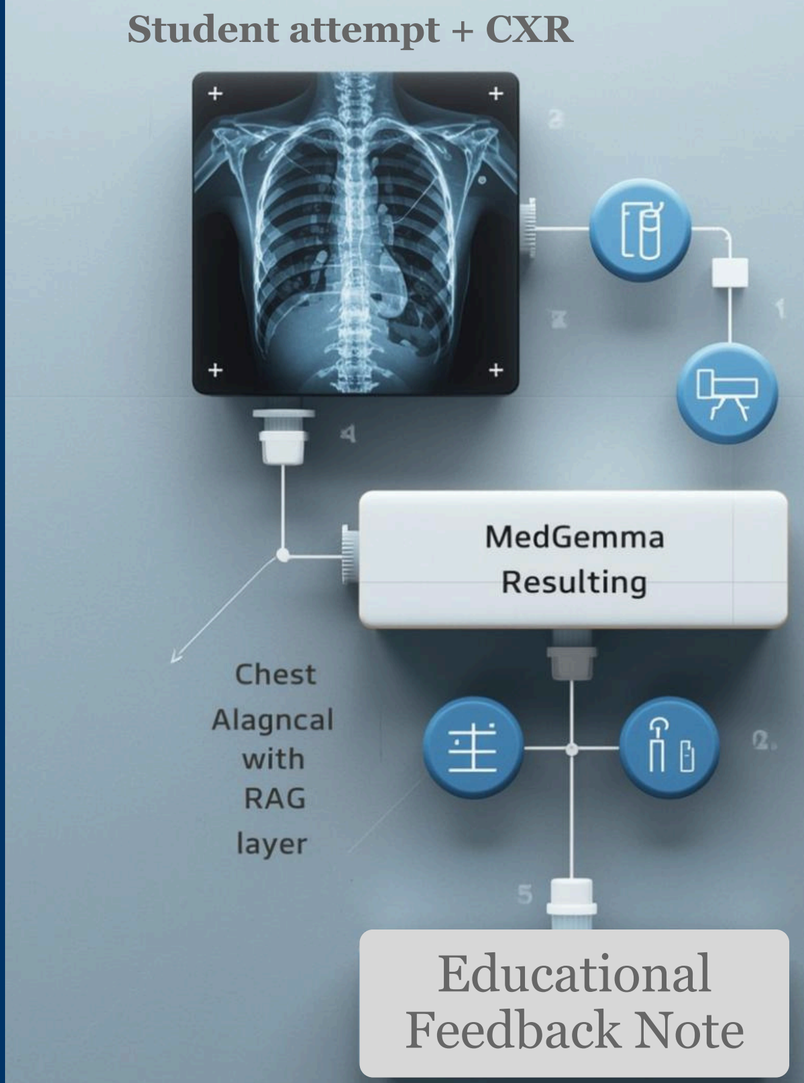
USING MEDGEMMA AND RETRIEVAL-AUGMENTED GENERATION FOR RADIOLOGY EDUCATION

The proposed system helps students practise CXR interpretation, compare their answer with an AI-assisted reference explanation, and receive feedback supported by retrieved medical evidence.

Proposed System Concept

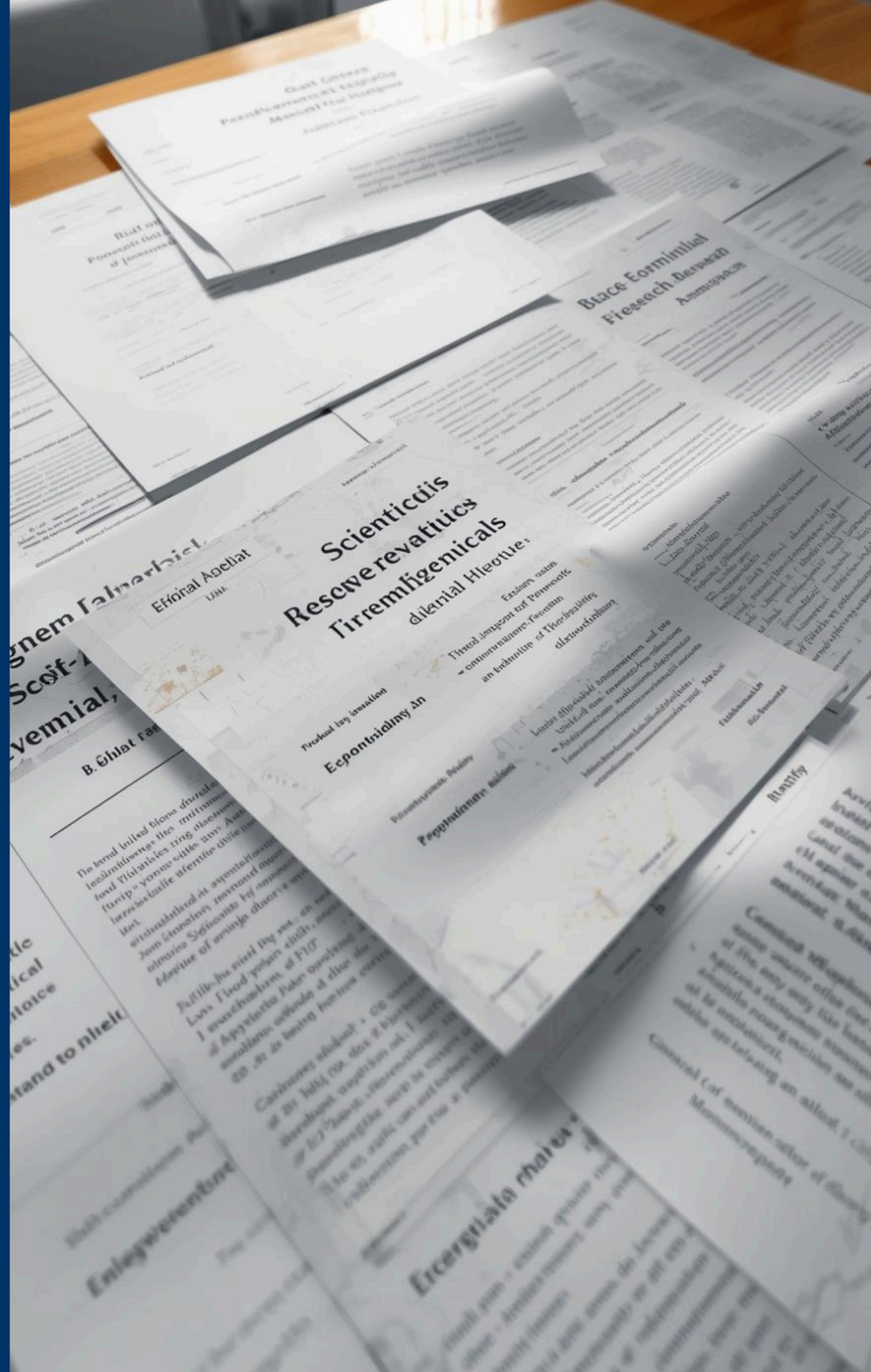
Suggested flow:

1. Student views chest X-ray
2. Student enters their own interpretation
3. MedGemma analyses the image
4. RAG retrieves relevant teaching evidence
5. System compares student answer with reference explanation
6. Student receives feedback, missed clues, and learning points



Literature Review

Ref	Data pre-process	RAG	Retrieval	Prompt eng.	Fine Tuning	CXR	Eval
1	✓	✗	✗	✗	✗	✓	✓
2	✓	✓	✓	✓	✗	✗	✓
3	✗	✗	✓	✓	✗	✓	✓
4	✓	✓	✓	✓	✗	✗	✓
5	✗	✓	✓	✓	✗	✗	✓
6	✓	✓	✓	✓	✓	✗	✓
7	✓	✓	✓	✓	✗	✗	✓



Literature Review

Ref	Data pre-process	RAG	Retrieval	Prompt eng.	Fine Tuning	CXR	Eval
8	✓	✓	✓	✗	✗	✓	✗
9	✓	✗	✗	✗	✓	✓	✓
10	✓	✗	✗	✓	✗	✓	✓
11	✓	✓	✓	✓	✗	✗	✓
12	✓	✓	✓	✓	✗	✗	✓
13	✓	✓	✓	✓	✗	✗	✓
14	✓	✓	✓	✓	✓	✗	✓



Literature Review

Ref	Data pre-process	RAG	Retrieval	Prompt eng.	Fine Tuning	CXR	Eval
15	✓	✗	✗	✗	✓	✓	✓
16	✗	✓	✓	✗	✓	✗	✗
17	✗	✗	✗	✓	✓	✗	✓
18	✓	✗	✗	✓	✗	✓	✓
19	✓	✗	✗	✓	✓	✓	✓
20	✓	✗	✗	✓	✓	✗	✓
21	✓	✗	✓	✗	✓	✓	✓
22	✓	✗	✗	✓	✗	✓	✓



Key Insights

- RAG INSIGHT: RAG CAN IMPROVE EVIDENCE GROUNDING AND EXPLANATION QUALITY
- MULTIMODAL CHALLENGE: CXR INTERPRETATION REQUIRES BOTH IMAGE UNDERSTANDING AND TEXTUAL EXPLANATION
- LACK OF EDUCATIONAL FOCUS: LIMITED FOCUS ON STUDENT-FACING RADIOLOGY TRAINING SYSTEMS

Research Gaps

- EVIDENCE GAP: LIMITED EVALUATION OF RETRIEVED EVIDENCE QUALITY IN FEEDBACK
- METHODOLOGICAL GAP: FEW COMPARISONS BETWEEN MEDGEMMA ONLY, MEDGEMMA + RAG, AND FINE-TUNED + RAG
- KNOWLEDGE GAP: NO CLEAR BEST PRACTICE FOR COMBINING MEDICAL VLMS WITH RAG FOR CXR TRAINING



Research Questions on Student Radiology Training

RQ1: MEDGEMMA ADAPTATION

How can MedGemma be adapted or prompted to support student radiologists in practising chest X-ray interpretation for selected findings?

RQ2: EVIDENCE RETRIEVAL IMPACT

Does RAG improve the grounding, clarity, and educational usefulness of feedback generated for chest X-ray training?

RQ3: OPTIMAL SETUP

Which setup provides the best balance between finding recognition, evidence grounding, and feedback quality: MedGemma only, MedGemma + RAG, or fine-tuned MedGemma + RAG?



Research Aims for Student Radiology Training

RA1: DESIGN PROTOTYPE

Develop a student-facing chest X-ray training companion using MedGemma and RAG.

RA2: EVALUATE FEEDBACK QUALITY

Assess whether RAG improves evidence grounding, explanation clarity, and usefulness for learning.

RA3: EDUCATIONAL POSITIONING

Position the system as a safe educational tool for radiology training.



Research Objectives

RO1: USE CASE SELECTION

Select a narrow set of CXR findings suitable for student training.

RO2: STUDENT WORKFLOW DEVELOPMENT

Design a workflow where students attempt interpretation before receiving AI feedback.

RO3: CORPUS CREATION

Create a focused educational and medical evidence corpus for RAG.

RO4: RAG IMPLEMENTATION

Implement retrieval and feedback generation based on selected findings.



Methodology Overview for Training Companion Evaluation

- STUDENT–IN–THE–LOOP PROTOTYPE INVESTIGATION
- COMPARATIVE EVALUATION OF MEDGEMMA ONLY VS MEDGEMMA + RAG
- PUBLIC CXR DATASETS AND SELECTED FINDINGS
- EVIDENCE–GROUNDED FEEDBACK GENERATION
- FEEDBACK QUALITY ASSESSMENT USING AN EDUCATIONAL RUBRIC



Research Methodology

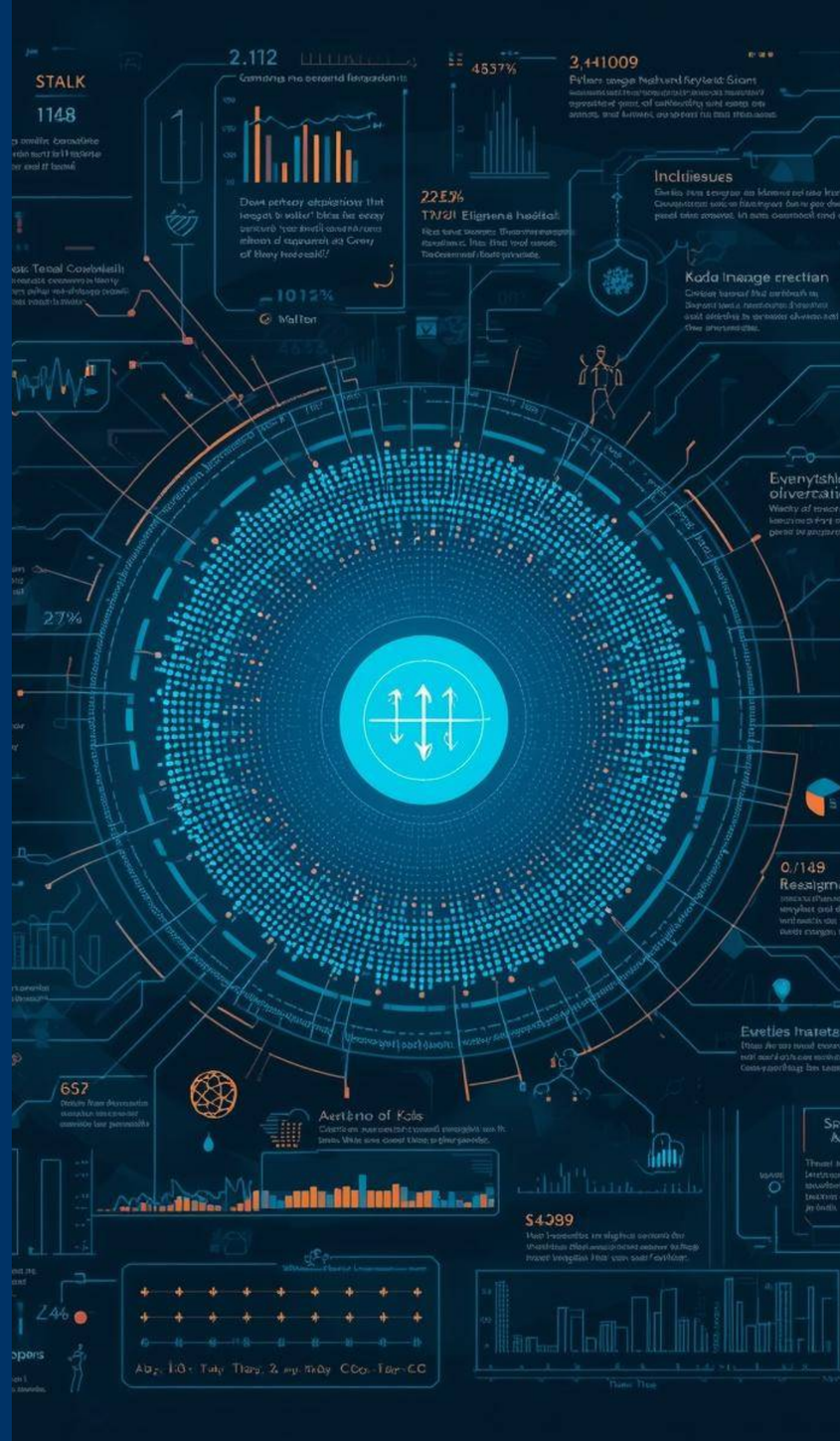
- LEARNING REQUIREMENTS DEFINITION: DEFINE WHAT FEEDBACK STUDENT RADIOLOGISTS NEED
- DATA PREPARATION: SELECT CXR IMAGES AND LABELS FOR TARGET FINDINGS
- KNOWLEDGE BASE CREATION: BUILD A FOCUSED EDUCATIONAL RAG CORPUS
- PROTOTYPE DEVELOPMENT: IMPLEMENT STUDENT ATTEMPT + AI FEEDBACK WORKFLOW
- TESTING AND EVALUATION: ASSESS TECHNICAL PERFORMANCE AND FEEDBACK USEFULNESS

Project Methodology Overview

CXR CASE → STUDENT INTERPRETATION → MEDGEMMA ANALYSIS → RAG RETRIEVAL → FEEDBACK GENERATION → STUDENT REFLECTION

Planned feedback components:

- Likely finding
- Key radiographic signs
- Comparison with student answer
- Missed or incorrect reasoning
- Retrieved supporting evidence
- Learning point or mini-quiz question



Datasets and Tools

- DATASETS USED: NIH CHEST X-RAY, MIMIC-CXR, OPTIONAL MIMIC-CXR-JPG
- TOOLS EMPLOYED: MEDGEMMA, RAG PIPELINE, VECTOR DATABASE, PROMPT ENGINEERING
- EVALUATION METRICS: FINDING RECOGNITION, FEEDBACK RELEVANCE, GROUNDING, CLARITY
- FEEDBACK USEFULNESS
- RETRIEVAL RELEVANCE



Expected Outcomes and Contributions

- DEMONSTRABLE PROTOTYPE: A STUDENT-FACING CXR TRAINING COMPANION USING MEDGEMMA + RAG
- EDUCATIONAL FEEDBACK FRAMEWORK: A WORKFLOW FOR GENERATING EVIDENCE-GROUNDED FEEDBACK FOR STUDENT RADIOLOGISTS
- COMPARATIVE EVALUATION: EVIDENCE ON WHETHER RAG IMPROVES FEEDBACK QUALITY OVER MEDGEMMA ALONE
- SAFE AI POSITIONING: A CLEAR EDUCATIONAL-USE FRAMING THAT AVOIDS CLINICAL DEPLOYMENT CLAIMS



Reference List

1. Pal, A., Lee, J.-O., Zhang, X., Sankarasubbu, M., Roh, S., Kim, W.J., Lee, M. and Rajpurkar, P. (2025) ReXVQA: A Large-scale Visual Question Answering Benchmark for Generalist Chest X-ray Understanding.
2. Papageorgiou, P.S., Christodoulou, R.C., Pitsillos, R. and Petrou, V. (2025) 'The Role of Large Language Models in Improving Diagnostic-Related Groups Assignment and Clinical Decision Support in Healthcare Systems: An Example from Radiology and Nuclear Medicine', Applied Sciences, 15(16), 9005.
3. Lou, J., Yang, Y., Yu, Z., Fu, Z., Han, W., Huang, Q. and Yu, J. (2025) CXRAgent: Director-Orchestrated Multi-Stage Reasoning for Chest X-Ray Interpretation. arXiv:2510.21324.
4. Mugambiwa, T. and Ndlovu, B. (2026) 'Multi-Agent Retrieval Augmented Generation for Clinical Decision Support: A Systematic Review and Integrative Conceptual Framework', Journal of Applied Informatics and Computing, 10(1).
5. IEEE Xplore document 11176078
6. Enhancing Clinical Decision Support with LLMs: A Feasibility Study Integrating CoT, RAG, and QLoRA (2025) Procedia Computer Science, 270, pp. 4946–4956.
7. Pambudi, S. and Menolascina, F. (2025) Bridging Clinical Narratives and ACR Appropriateness Guidelines: A Multi-Agent RAG System for Medical Imaging Decisions. arXiv:2510.04969.
8. IEEE Xplore document 11322500
9. Lee, S., Youn, J., Kim, H., Kim, M. and Yoon, S.H. (2025) 'CXR-LLaVA: a multimodal large language model for interpreting chest X-ray images', European Radiology, 35(7), pp. 4374–4386.
10. ChestX-VQA: AI Tool for Multimodal Chest X-ray Analysis and Clinical QA (preprint; user-provided source).
11. Liu, S., McCoy, A.B. and Wright, A. (2025) 'Improving large language model applications in biomedicine with retrieval-augmented generation: a systematic review, meta-analysis, and clinical development guidelines', Journal of the American Medical Informatics Association, 32(4), pp. 605–615.



Reference List

12. Bahraini, A. et al. (2025) ‘Retrieval augmented generation for large language models in healthcare: A systematic review’, PLOS Digital Health.
13. Abo El-Enen, M., Saad, S. and Nazmy, T. (2025) ‘A survey on retrieval-augmentation generation (RAG) models for healthcare applications’, Neural Computing and Applications.
14. Collaco, B.G. et al. (2026) ‘Integrating Fine-Tuning and Retrieval-Augmented Generation for Healthcare AI Systems: A Scoping Review’, Bioengineering, 13(2), 225.
15. Sellergren, A. et al. (2025) MedGemma Technical Report. arXiv:2507.05201.
16. HAL thesis record tel-05481134 (user-provided source; complete bibliographic metadata to verify).
17. Lin, Q. et al. (2025) Toward Clinically Explainable AI for Medical Diagnosis: A Foundation Model with Human-Compatible Reasoning via Reinforcement Learning. arXiv:2509.03906.
18. Gosai, A. et al. (2025) Beyond Diagnosis: Evaluating Multimodal LLMs for Pathology Localization in Chest Radiographs. arXiv:2509.18015.
19. Chen, Z. et al. (2024) CheXagent: Towards a Foundation Model for Chest X-Ray Interpretation. arXiv:2401.12208.
20. Li, C. et al. (2023) LLaVA-Med: Training a Large Language-and-Vision Assistant for Biomedicine in One Day. Advances in Neural Information Processing Systems 36.
21. Zambrano Chaves, J.M. et al. (2025) ‘A clinically accessible small multimodal radiology model and evaluation metric for chest X-ray findings’, Nature Communications, 16, 3108.
22. MedRAX: AI Chest X-Ray Agent / MedRAX chest X-ray analysis framework (project and paper landing pages; bibliographic metadata may vary by release).



Student-in-the-Loop Training Companion

This research proposal outlines a student-in-the-loop chest X-ray training companion using MedGemma and Retrieval-Augmented Generation. The system aims to support radiology students by providing structured, evidence-grounded feedback for selected CXR findings, while remaining clearly positioned as an educational tool rather than a clinical decision support system.

